FOUR EQUIVALENT FORMULATIONS OF THE J-2 PROPERTY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Editorial note. Benign-screen note: the proof contains the phrase 'it is clear that' / 'is obvious' once, used as a legitimate step inside a complete argument; the author proves the statement, so this is not a hint-only proof.

Allowed standard prerequisites (editorial).

- A J-1 domain is J-0, and the criterion for J-2 through quotients by primes. Lemma \ref{lemma-intersection-regular-with-closed}, cited at proof lines 11710-11712. Give two of the implications.
- Existence of a diagram of finite purely inseparable extensions making the resulting extension separable. Algebra, Lemma \ref{algebra-lemma-make-separably-generated}, cited at proof lines 11722-11724. Produces the extension used in the final step.
- Ascent and descent of the J-0 property. Lemma \ref{lemma-J-0-goes-up} and Lemma \ref{lemma-J-0-goes-down}, cited at proof lines 11731 and 11732-11734. Transfer J-0 between the auxiliary domain and the original one.

Source reference. Original lemma 07PC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9792–9792. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 07P9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9789–9789. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 04KM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5978–5978. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07PB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9791–9791. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07PA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9790–9790. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a Noetherian ring. The following are equivalent*

(1) *R is J-2,*
(2) *every finite type R-algebra which is a domain is J-0,*
(3) *every finite R-algebra is J-1,*
(4) *for every prime $\mathfrak{p}$ and every finite purely inseparable extension $L/\kappa(\mathfrak{p})$ there exists a finite R-algebra R' which is a domain, which is J-0, and whose field of fractions is L.*

Proof. It is clear that we have the implications (1) $\Rightarrow$ (2) and (2) $\Rightarrow$ (4). Recall that a domain which is J-1 is J-0. Hence we also have the implications (1) $\Rightarrow$ (3) and (3) $\Rightarrow$ (4).

Let $R \to S$ be a finite type ring map and let's try to show S is J-1. By Lemma 07P9 it suffices to prove that $S/\mathfrak{q}$ is J-0 for every prime $\mathfrak{q}$ of S. In this way we see (2) $\Rightarrow$ (1).

Assume (4). We will show that (2) holds which will finish the proof. Let $R \to S$ be a finite type ring map with S a domain. Let $\mathfrak{p} = \mathrm{Ker}(R \to S)$. Let K be the fraction field of S. There exists a diagram of fields

$$\begin{array}{ccc} K & \longrightarrow & K' \\ \uparrow & & \uparrow \\ \kappa(\mathfrak{p}) & \longrightarrow & L \end{array}$$

where the horizontal arrows are finite purely inseparable field extensions and where K'/L is separable, see Algebra, Lemma 04KM. Choose $R' \subset L$ as in (4) and let S' be the image of the map $S \otimes_R R' \to K'$. Then S' is a domain whose fraction field is K', hence S' is J-0 by Lemma 07PB and our choice of R'. Then we apply Lemma 07PA to see that S is J-0 as desired. $\square$